Parameter Sensitivity and Computational Analysis

This section evaluates the effects of the neighborhood sample count n and sample spacing h on the direct squared-loss, iterative squared-loss, and correntropy gradient estimators. The implementation applies a separable one-dimensional Taylor approximation independently along the horizontal and vertical axes, with polynomial order $p = 3$. The parameter n denotes the number of symmetric samples per axis, whereas h specifies the distance between successive samples. All gradient RMSE and HOG metrics are computed relative to the clean output obtained with the same estimator and parameter setting. They therefore measure *noise stability*, rather than absolute derivative accuracy. Gaussian noise at 20 dB is reported as the representative distributed-noise condition. Clean matched-self rows are omitted because they trivially give RMSE $= 0$, HOG error $= 0$, and cosine similarity $= 1$.

Influence of the sample count. The n sensitivity experiment fixes $h = 1$ and evaluates $n \in \{4, 8, 16, 32\}$. Increasing n substantially reduces gradient RMSE for all three methods. Under Gaussian noise at 20 dB, for example, the direct squared-loss RMSE decreases from 7.077 at $n = 4$ to 0.302 at $n = 32$. Nevertheless, the HOG relative error does not improve monotonically: for the same method it changes from 0.3527 at $n = 8$ to 0.3671 at $n = 32$. Thus, a larger support improves gradient-magnitude stability but may reduce spatial locality and alter the orientation distribution used by HOG.

Under salt-and-pepper noise, correntropy benefits markedly from larger neighborhoods. At 5% noise and $n = 16$, it reduces RMSE from 4.686 to 2.262 and HOG relative error from 0.5535 to 0.2526 relative to direct squared loss, while increasing cosine similarity from 0.8468 to 0.9681. At $n = 32$, the corresponding correntropy values improve to 0.899, 0.2027, and 0.9795. A similar advantage is observed at 10% noise. The identical direct and correntropy results observed in several $n = 4$ cases should be interpreted carefully: with $p = 3$ and only four samples, the system has limited redundancy for robust reweighting.

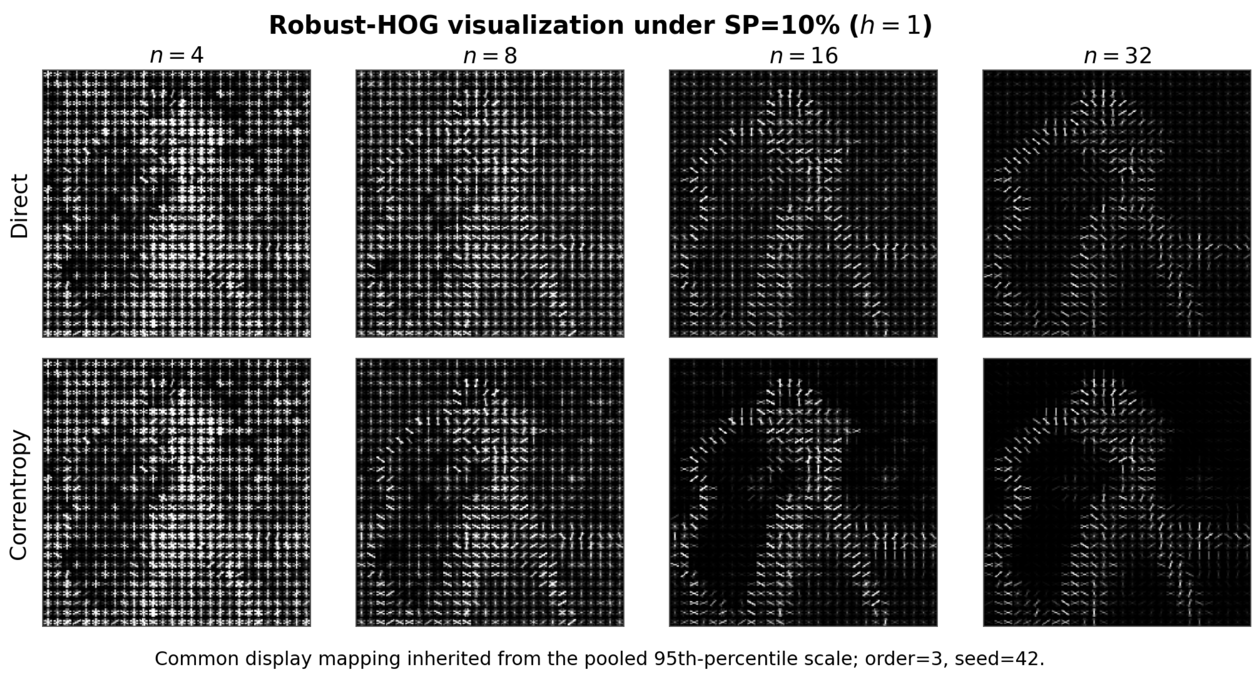

Figure 1: Qualitative Robust-HOG comparison under 10% salt-and-pepper noise for different values of n, with $h = 1$ and $p = 3$.

Influence of sample spacing. The h sensitivity experiment fixes $n = 8$ and evaluates $h \in \{1, 2, 3, 4\}$. Under Gaussian noise, increasing h reduces gradient RMSE but slightly degrades the HOG metrics. For direct squared loss, RMSE decreases from 2.413 at $h = 1$ to 0.573 at $h = 4$, whereas HOG relative error increases from 0.3527 to 0.3896. This indicates stronger smoothing without a corresponding improvement in descriptor stability.

The opposite tendency is observed under impulsive noise. At 5% salt-and-pepper noise, increasing the correntropy spacing from $h = 1$ to $h = 4$ reduces RMSE from 8.196 to 2.288, reduces HOG relative error from 0.4762 to 0.4120, and increases cosine similarity from 0.8866 to 0.9151. Similar improvements occur at 10% noise. Hence, $h = 4$ provides the strongest tested stability under salt-and-pepper noise, although it is the boundary of the evaluated range and is not claimed to be globally optimal.

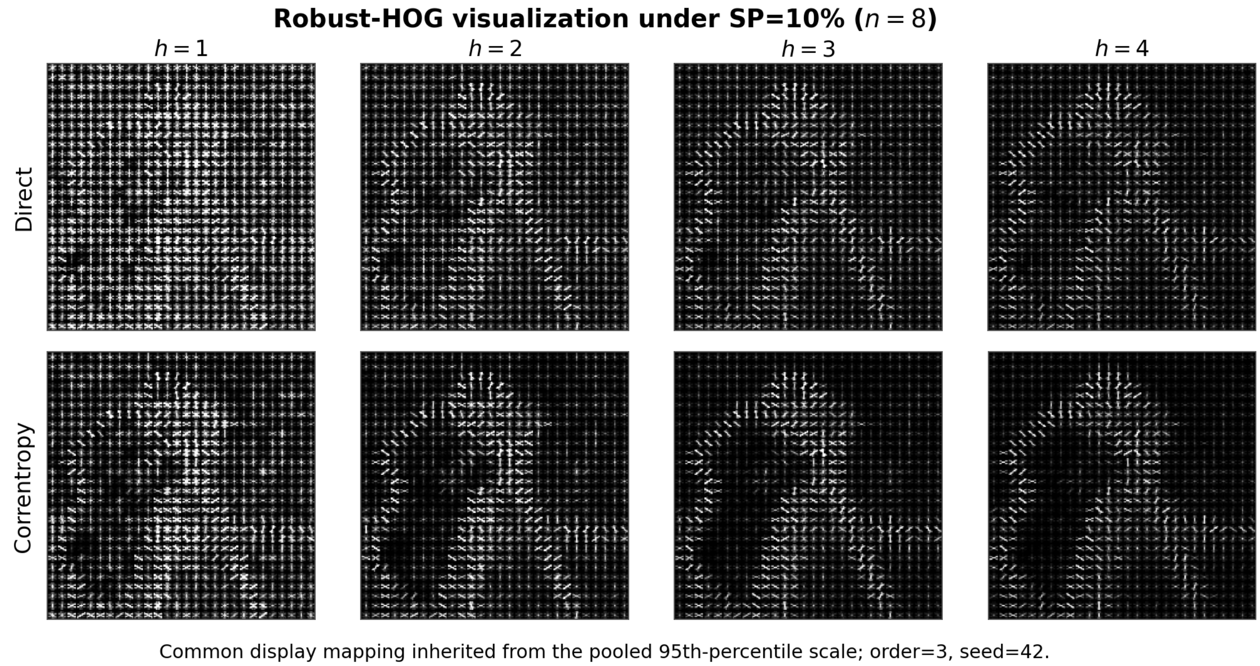

Figure 2: Qualitative Robust-HOG comparison under 10% salt-and-pepper noise for different values of h, with $n = 8$ and $p = 3$.

Comparison of the three estimators. Tables 1 and 2 report the matched-setting results for Gaussian noise at 20 dB and salt-and-pepper noise at 5% and 10%. Gaussian 20 dB is used as the representative distributed-noise condition. Within each noise and parameter setting, the three estimators are listed consecutively and use the same image, region of interest, HOG settings, and seed-42 noise realization. Clean matched-self rows are omitted because they trivially give RMSE = 0, HOG error = 0, and cosine similarity = 1.

Table 1: Neighborhood-count sensitivity at fixed $h = 1$ for Gaussian 20 dB and salt-and-pepper noise at 5% and 10%. Rows are ordered by noise, n, and method.

Noise	n	Method	Matched-clean Gradient RMSE	HOG Relative Error	Cosine Similarity	Runtime (s)
Gaussian 20 dB	4	Direct squared loss	7.077	0.3895	0.9241	0.222
Gaussian 20 dB	4	Iterative squared loss	7.707	0.3900	0.9239	0.655
Gaussian 20 dB	4	Correntropy loss	7.077	0.3895	0.9241	0.456
Gaussian 20 dB	8	Direct squared loss	2.413	0.3527	0.9378	0.226
Gaussian 20 dB	8	Iterative squared loss	2.342	0.3523	0.9379	0.927
Gaussian 20 dB	8	Correntropy loss	3.023	0.3581	0.9359	26.098
Gaussian 20 dB	16	Direct squared loss	0.859	0.3635	0.9339	0.254
Gaussian 20 dB	16	Iterative squared loss	0.834	0.3644	0.9336	1.038
Gaussian 20 dB	16	Correntropy loss	1.170	0.3688	0.9320	21.947
Gaussian 20 dB	32	Direct squared loss	0.302	0.3671	0.9326	0.305
Gaussian 20 dB	32	Iterative squared loss	0.294	0.3673	0.9325	1.481
Gaussian 20 dB	32	Correntropy loss	0.491	0.3942	0.9223	21.497
SP 5%	4	Direct squared loss	40.064	0.5427	0.8527	0.207
SP 5%	4	Iterative squared loss	43.623	0.5543	0.8464	0.634
SP 5%	4	Correntropy loss	40.064	0.5427	0.8527	0.416
SP 5%	8	Direct squared loss	13.374	0.6077	0.8153	0.220
SP 5%	8	Iterative squared loss	12.980	0.6071	0.8157	0.875
SP 5%	8	Correntropy loss	8.196	0.4762	0.8866	18.011
SP 5%	16	Direct squared loss	4.686	0.5535	0.8468	0.245
SP 5%	16	Iterative squared loss	4.552	0.5538	0.8466	0.980
SP 5%	16	Correntropy loss	2.262	0.2526	0.9681	13.015
SP 5%	32	Direct squared loss	1.618	0.5563	0.8453	0.293
SP 5%	32	Iterative squared loss	1.572	0.5567	0.8451	1.378
SP 5%	32	Correntropy loss	0.899	0.2027	0.9795	20.071

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Table 1 (continued)

Noise	n	Method	Matched-clean Gradient RMSE	HOG Relative Error	Cosine Similarity	Runtime (s)
SP 10%	4	Direct squared loss	56.435	0.6633	0.7800	0.208
SP 10%	4	Iterative squared loss	61.458	0.6675	0.7772	0.633
SP 10%	4	Correntropy loss	56.435	0.6633	0.7800	0.415
SP 10%	8	Direct squared loss	18.820	0.6316	0.8005	0.229
SP 10%	8	Iterative squared loss	18.265	0.6315	0.8006	0.874
SP 10%	8	Correntropy loss	13.362	0.5659	0.8399	21.340
SP 10%	16	Direct squared loss	6.663	0.5896	0.8262	0.245
SP 10%	16	Iterative squared loss	6.473	0.5899	0.8260	0.990
SP 10%	16	Correntropy loss	3.942	0.4333	0.9061	17.735
SP 10%	32	Direct squared loss	2.323	0.6135	0.8118	0.289
SP 10%	32	Iterative squared loss	2.257	0.6139	0.8115	1.377
SP 10%	32	Correntropy loss	1.368	0.3321	0.9449	25.201

All rows use $p = 3$, seed 42, the same 224×224 ROI, and identical estimator and HOG settings. Runtime is the median of three gradient-plus-HOG runs; solver setup and matched-clean reference computation are excluded. Gaussian 20 dB is the representative distributed-noise condition. Clean matched-self rows are omitted because they trivially give RMSE = 0, HOG error = 0, and cosine similarity = 1.

Table 2: Sample-spacing sensitivity at fixed $n = 8$ for Gaussian 20 dB and salt-and-pepper noise at 5% and 10%. Rows are ordered by noise, h, and method.

Noise	h	Method	Matched-clean Gradient RMSE	HOG Relative Error	Cosine Similarity	Runtime (s)
Gaussian 20 dB	1	Direct squared loss	2.413	0.3527	0.9378	0.226
Gaussian 20 dB	1	Iterative squared loss	2.342	0.3523	0.9379	0.927
Gaussian 20 dB	1	Correntropy loss	3.023	0.3581	0.9359	26.098
Gaussian 20 dB	2	Direct squared loss	1.179	0.3716	0.9309	0.229
Gaussian 20 dB	2	Iterative squared loss	1.145	0.3726	0.9306	0.932
Gaussian 20 dB	2	Correntropy loss	1.834	0.3734	0.9303	16.145
Gaussian 20 dB	3	Direct squared loss	0.770	0.3856	0.9256	0.227
Gaussian 20 dB	3	Iterative squared loss	0.747	0.3867	0.9252	0.879
Gaussian 20 dB	3	Correntropy loss	1.200	0.3895	0.9241	16.309
Gaussian 20 dB	4	Direct squared loss	0.573	0.3896	0.9241	0.225
Gaussian 20 dB	4	Iterative squared loss	0.556	0.3906	0.9237	0.835
Gaussian 20 dB	4	Correntropy loss	1.034	0.3957	0.9217	17.647
SP 5%	1	Direct squared loss	13.374	0.6077	0.8153	0.220
SP 5%	1	Iterative squared loss	12.980	0.6071	0.8157	0.875
SP 5%	1	Correntropy loss	8.196	0.4762	0.8866	18.011
SP 5%	2	Direct squared loss	6.453	0.5617	0.8423	0.220
SP 5%	2	Iterative squared loss	6.263	0.5617	0.8422	0.824
SP 5%	2	Correntropy loss	4.332	0.4431	0.9018	12.785
SP 5%	3	Direct squared loss	4.190	0.5310	0.8590	0.215
SP 5%	3	Iterative squared loss	4.067	0.5313	0.8589	0.801
SP 5%	3	Correntropy loss	2.918	0.4237	0.9102	11.862
SP 5%	4	Direct squared loss	3.078	0.5134	0.8682	0.212
SP 5%	4	Iterative squared loss	2.987	0.5133	0.8683	0.759
SP 5%	4	Correntropy loss	2.288	0.4120	0.9151	13.706
SP 10%	1	Direct squared loss	18.820	0.6316	0.8005	0.229
SP 10%	1	Iterative squared loss	18.265	0.6315	0.8006	0.874
SP 10%	1	Correntropy loss	13.362	0.5659	0.8399	21.340
SP 10%	2	Direct squared loss	9.123	0.6081	0.8151	0.222
SP 10%	2	Iterative squared loss	8.855	0.6085	0.8149	0.900
SP 10%	2	Correntropy loss	6.860	0.5413	0.8535	15.682
SP 10%	3	Direct squared loss	5.927	0.5984	0.8210	0.215
SP 10%	3	Iterative squared loss	5.752	0.5989	0.8207	0.854
SP 10%	3	Correntropy loss	4.588	0.5341	0.8574	11.788
SP 10%	4	Direct squared loss	4.331	0.5933	0.8240	0.213
SP 10%	4	Iterative squared loss	4.203	0.5937	0.8237	0.791
SP 10%	4	Correntropy loss	3.486	0.5243	0.8626	8.903

All rows use $p = 3$, seed 42, the same 224×224 ROI, and identical estimator and HOG settings. Runtime is the median of three gradient-plus-HOG runs; solver setup and matched-clean reference computation are excluded. Gaussian 20 dB is the representative distributed-noise condition. Clean matched-self rows are omitted because they trivially give RMSE = 0, HOG error = 0, and cosine similarity = 1.

The direct and iterative squared-loss methods produce almost identical HOG metrics, while the iterative implementation is consistently several times slower. For instance, at $n = 16$ under 10% salt-and-pepper noise, their runtimes are 0.245 s and 0.990 s, although their HOG relative errors are 0.5896 and 0.5899. The iterative method therefore provides no substantial practical advantage in the present experiments.

Under the representative Gaussian 20 dB condition, the squared-loss estimators are generally slightly more accurate and considerably faster than correntropy. At $n = 8$ and $h = 1$, their RMSE values are 2.413, 2.342, and 3.023 for the direct, iterative, and correntropy methods, respectively. In contrast, correntropy provides a clear advantage under salt-and-pepper noise, particularly for $n = 16$ and $n = 32$. Its benefit should therefore be interpreted as robustness to impulsive outliers rather than universal superiority across noise models.

Computational cost. Figure 3 reports the median runtime of the complete gradient-plus-HOG pipeline. The direct estimator is consistently fastest, followed by the iterative squared-loss method, while correntropy is substantially more expensive because of its repeated data-dependent reweighting. For a 256×256 image with $n = 32$, the measured runtimes are 0.328 s, 1.641 s, and 28.081 s for the direct, iterative, and correntropy methods, respectively. Correntropy runtime also depends on convergence behavior and is therefore not expected to vary strictly monotonically with n.

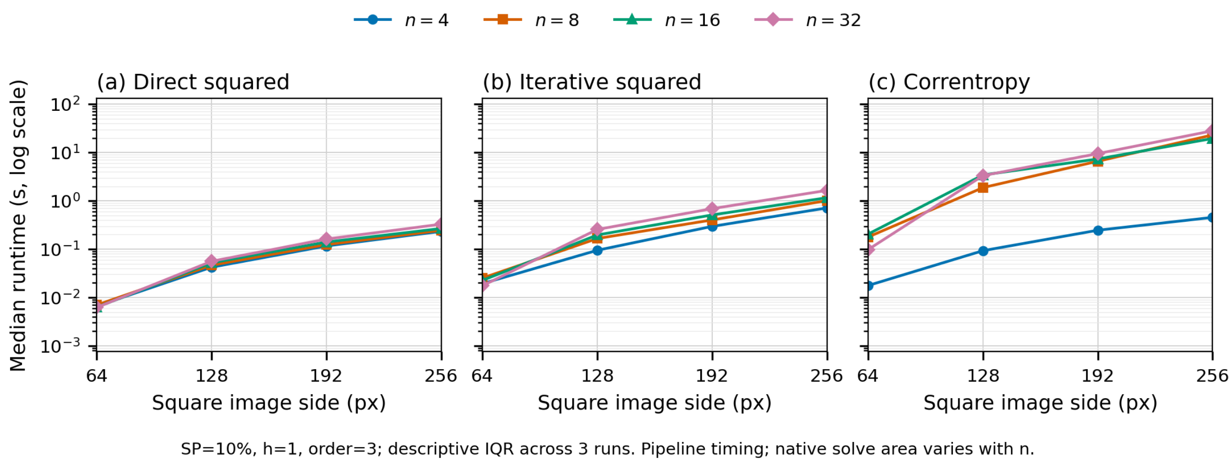

Figure 3: Median gradient-plus-HOG runtime under 10% salt-and-pepper noise for $h = 1$ and $p = 3$. The shared vertical axis is logarithmic, and shaded regions show interquartile ranges over three repetitions.

Recommended configurations. The results indicate that no single configuration is optimal for all noise models and computational constraints. Direct squared loss with $n = 4$ or $n = 8$ is the preferred efficiency-oriented configuration and remains competitive under Gaussian noise. The iterative squared-loss method produces similar quality at higher computational cost and is therefore retained mainly as an optimization baseline. Among the measured correntropy candidates, $(n, h) = (16, 4)$ provides the most practical balance between impulsive-noise stability and runtime, whereas $(n, h) = (32, 1)$ provides the strongest measured robustness at substantially higher cost.

Overall, squared loss is preferable for Gaussian noise and runtime-constrained applications, whereas correntropy is preferable when robustness to salt-and-pepper noise is the primary requirement. These conclusions are based on one source image and seed 42. Moreover, the n and h studies are one-factor-at-a-time experiments; therefore, no unmeasured combination, including $(32, 4)$, is claimed to be optimal.